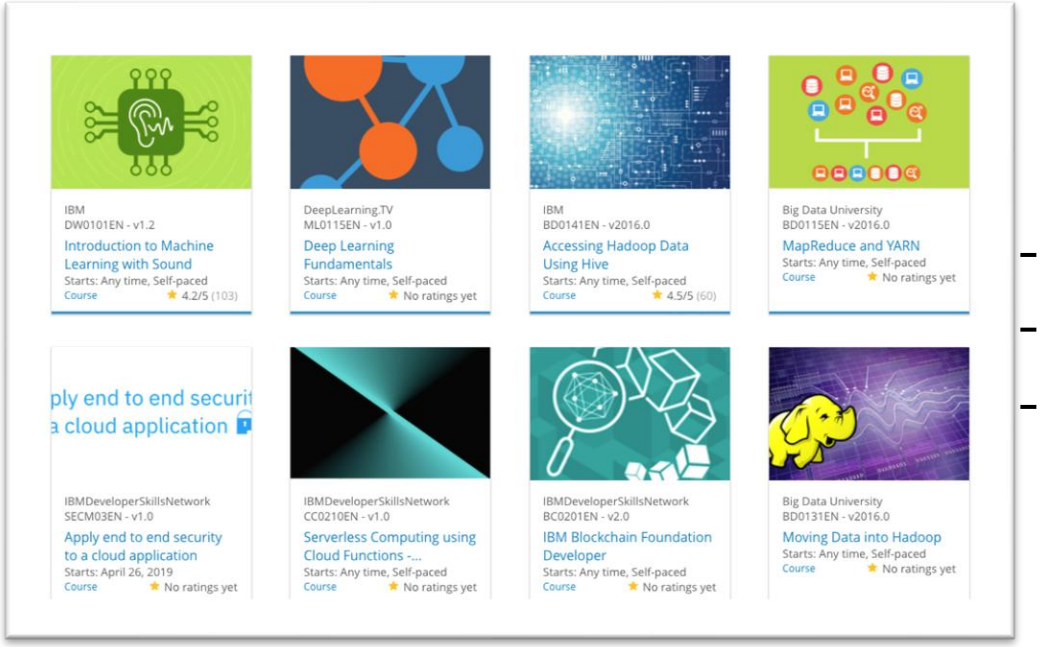
Build a Personalized Online Course Recommender System with Machine Learning

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IBM Machine Learning
Professional Certificate
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Outline



Introduction and Background



Exploratory Data Analysis



Content-based Recommender System using Unsupervised Learning



Collaborative-filtering based Recommender System using Supervised learning



Conclusion



Appendix

Project Background & Context

Background

- Online learning platforms offer hundreds of courses across multiple domains.
- Learners often struggle to identify courses that match their interests and learning goals.
- Recommendation systems help personalize learning experiences by suggesting relevant courses based on historical interactions.

Context

- Dataset consists of:
  - 233,306 course ratings
  - 33,901 users
  - 307 content-based courses
  - 126 courses used in collaborative-filtering models

Multiple recommendation approaches were evaluated:

- Content-Based Filtering
- Clustering-Based Filtering
- KNN Collaborative Filtering
- Matrix Factorization (NMF)
- Neural Collaborative Filtering
- Embedding-Based Regression
- Classification Models

Research Questions & Hypotheses

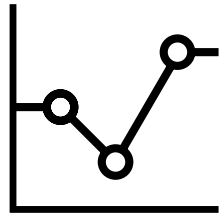
Research Questions

- Which recommendation approach produces the most effective course recommendations?
- Can content-based filtering generate meaningful recommendations from course features and user interests?
- Can clustering users improve recommendation quality?
- Which collaborative-filtering model achieves the lowest RMSE?
- How does dataset sparsity affect recommendation performance?

Hypotheses

- 1: Content-based recommenders can generate relevant personalized recommendations.
- 2: Users with similar learning interests can be grouped using clustering techniques.
- 3: Recommendation performance varies across algorithms and hyperparameter settings.
- 4: Neural network embeddings can learn meaningful latent user and course representations.
- 5: High sparsity negatively impacts matrix factorization approaches such as NMF.
- 6: Neural collaborative filtering will achieve the best overall prediction accuracy.

Exploratory Data Analysis



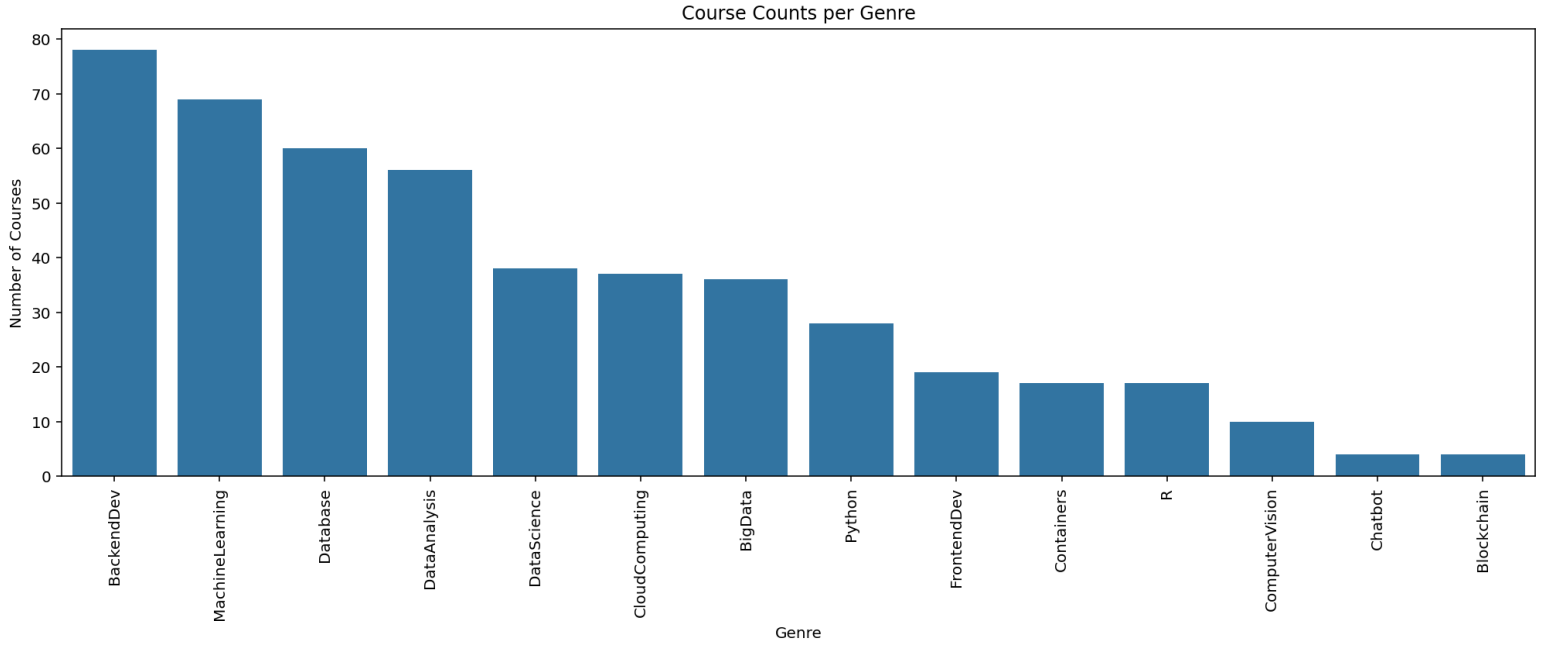
Course counts per genre

Key Findings

- Backend Development contains the most courses (**78**).
- Machine Learning is the second largest genre (**69**).
- Database (**60**) and Data Analysis (**56**) are also highly represented.
- Chatbot (**4**) and Blockchain (**4**) contain the fewest courses.

Insight

- The course catalog spans 14 genres, with the strongest representation in software development, data science, and machine learning. This provides a diverse foundation for building personalized recommendation systems.



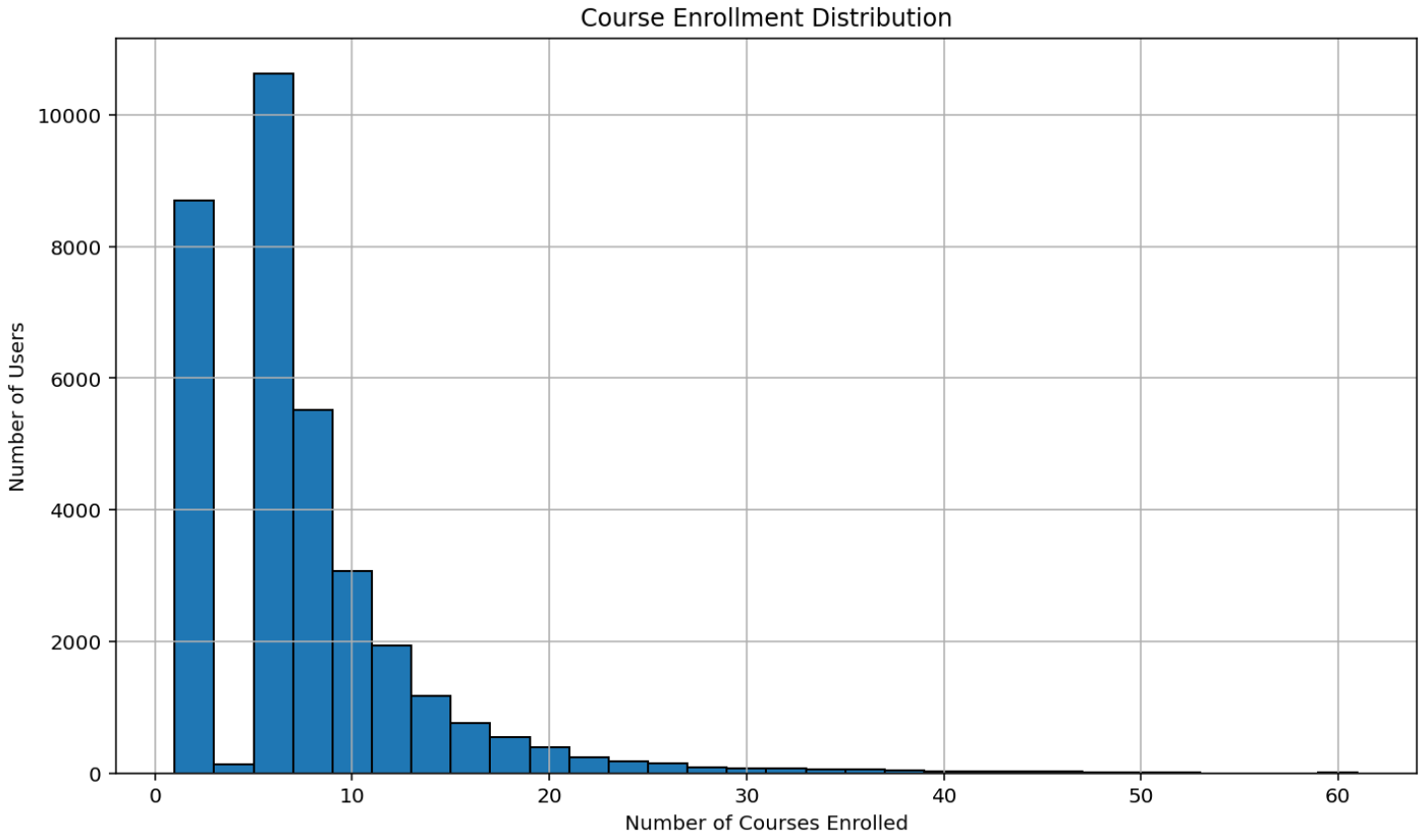
Course enrollment distribution

Key Findings

- Most users enrolled in fewer than 10 courses
- Average enrollments per user: 6.9
- Median enrollments per user: 6
- Maximum enrollments observed: 61
- Enrollment distribution is right-skewed.

Insight

- Most learners enroll in only a few courses, while a small number of highly active learners enroll in many courses. This variation in behavior highlights the need for personalized course recommendations.



20 most popular courses

Key Findings

- Python for Data Science was the most popular course (14,936 enrollments).
- Introduction to Data Science ranked second (14,477 enrollments).
- Big Data 101 ranked third (13,291 enrollments).
- Data science, machine learning, and big data courses dominated the rankings.
- The Top 20 courses accounted for approximately 63.3% of total enrollments.

Key Insight

- Learner demand is concentrated around a relatively small number of foundational data science, analytics, and machine learning courses.

Rank	Course Title	Enrollments
1	Python for Data Science	14,936
2	Introduction to Data Science	14,477
3	Big Data 101	13,291
4	Hadoop 101	10,599
5	Data Analysis with Python	8,303
6	Data Science Methodology	7,719
7	Machine Learning with Python	7,644
8	Spark Fundamentals I	7,551
9	Data Science Hands-on with Open Source Tools	7,199
10	Blockchain Essentials	6,719
11	Data Visualization with Python	6,709
12	Deep Learning 101	6,323
13	Build Your Own Chatbot	5,512
14	R for Data Science	5,237
15	Statistics 101	5,015
16	Introduction to Cloud	4,983
17	Docker Essentials: A Developer Introduction	4,480
18	SQL and Relational Databases 101	3,697
19	MapReduce and YARN	3,670
20	Data Privacy Fundamentals	3,624

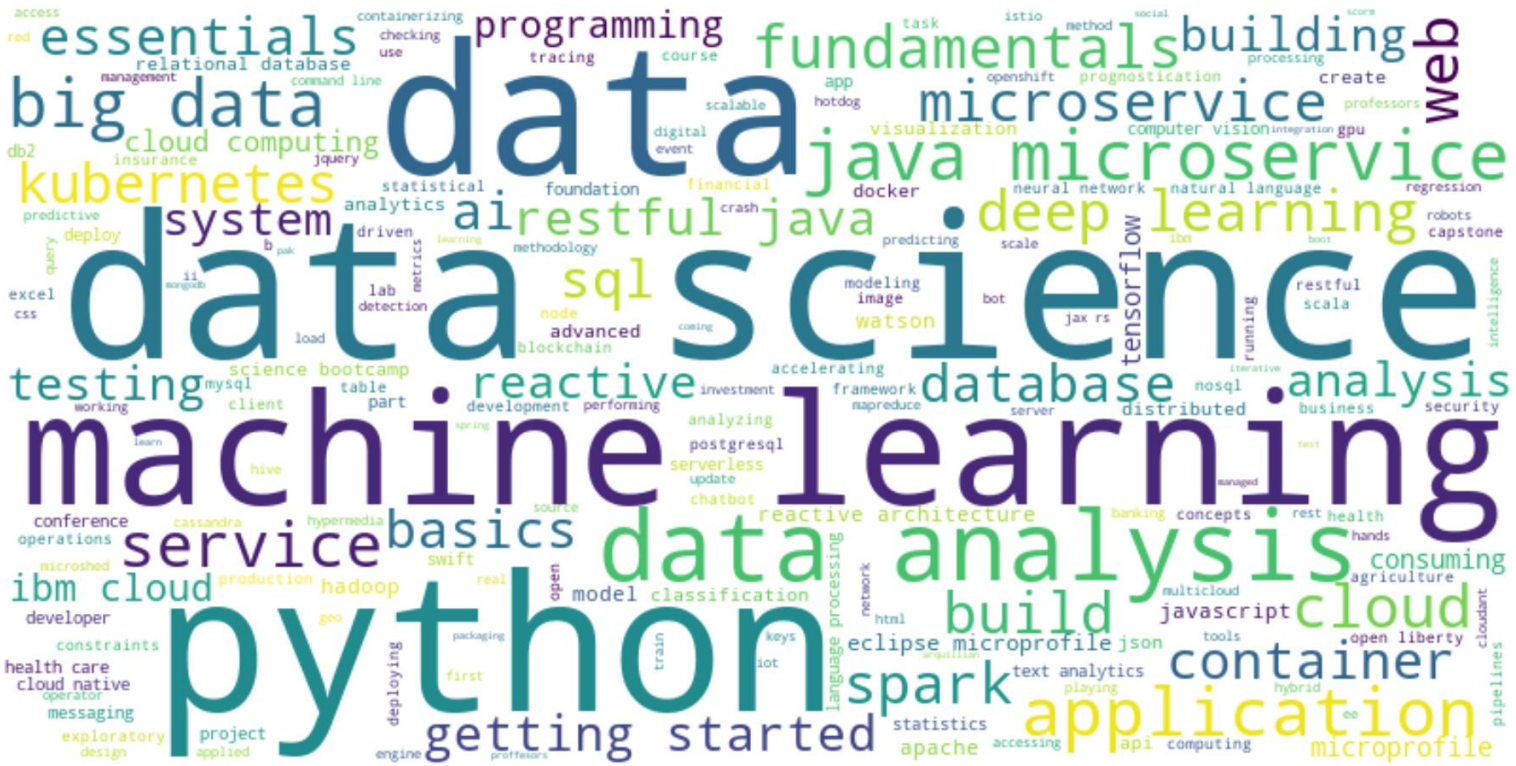
Word cloud of course titles

Key Findings

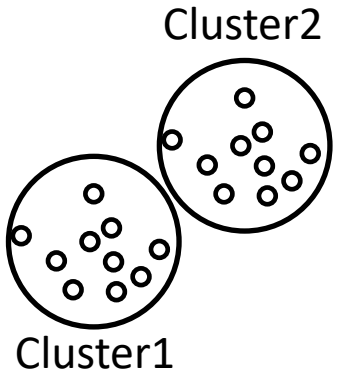
- "Data", "Science", and "Python" are the most prominent words.
- Machine Learning, Cloud, SQL, and Database are also frequently represented.
- Most course titles focus on data science, analytics, AI, and software development.
- The catalog is strongly aligned with current technology and data-driven skills.

Key Insight

- The course catalog is heavily concentrated in data science, machine learning, analytics, and cloud computing topics.



Content-based Recommender System using Unsupervised Learning



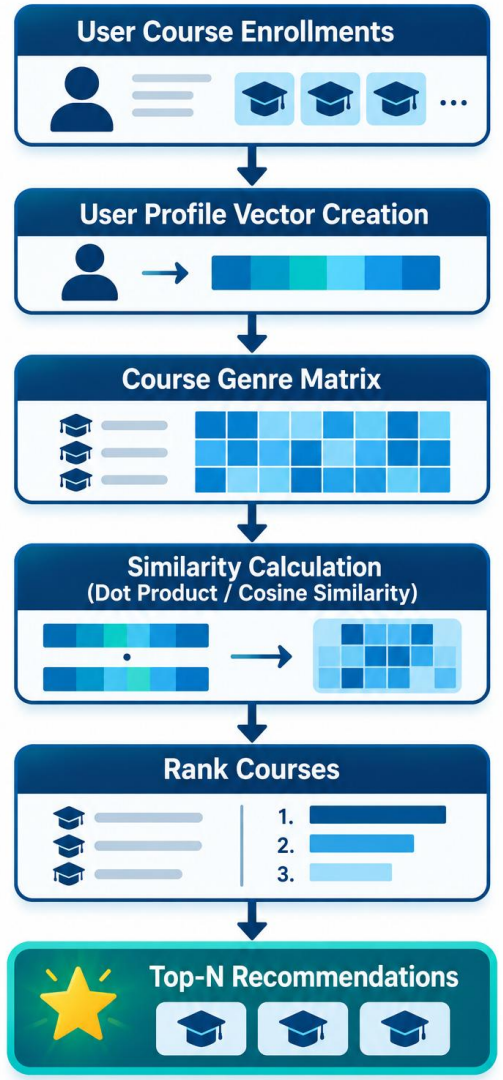
Flowchart of content-based recommender system using user profile and course genres

Key Steps

- Collect user enrollment history
- Build user profile from enrolled course genres
- Represent courses using genre vectors
- Calculate similarity scores
- Rank unseen courses
- Recommend highest-ranked courses
- Generate recommendation scores using dot product similarity

Key Insight

- Recommendations are generated using course content and user interests rather than learner-to-learner similarities.



Evaluation results of user profile-based recommender system

Hyperparameter Settings

- Recommendation Score Threshold = **10.0**
- User Profile Vector = **Course Genre Preferences**
- Recommendation Method = **Content-Based Filtering**

Recommendation Statistics

- Users Evaluated: **33,901**
- Total Recommendations: **1,500,424**
- Average Recommendations per User: **60.82**
- Average Recommendation Score: **19.12**

Key Finding

- Each user received approximately **61 unseen course recommendations** on average.

Course Title	Recommendation Count
Text Analytics at Scale	17,390
Applied Machine Learning in Python	15,656
Introduction to Data Science in Python	15,656
Data Science in Insurance	15,644
Accelerating Deep Learning with GPU	15,603
SQL for Data Science	15,062
SQL for Data Science Capstone Project	15,062
Database Operations in Cloudant	14,689
Analyzing Big Data with SQL	14,464
Foundations for Big Data Analysis with SQL	14,464

Flowchart of content-based recommender system using course similarity

Method

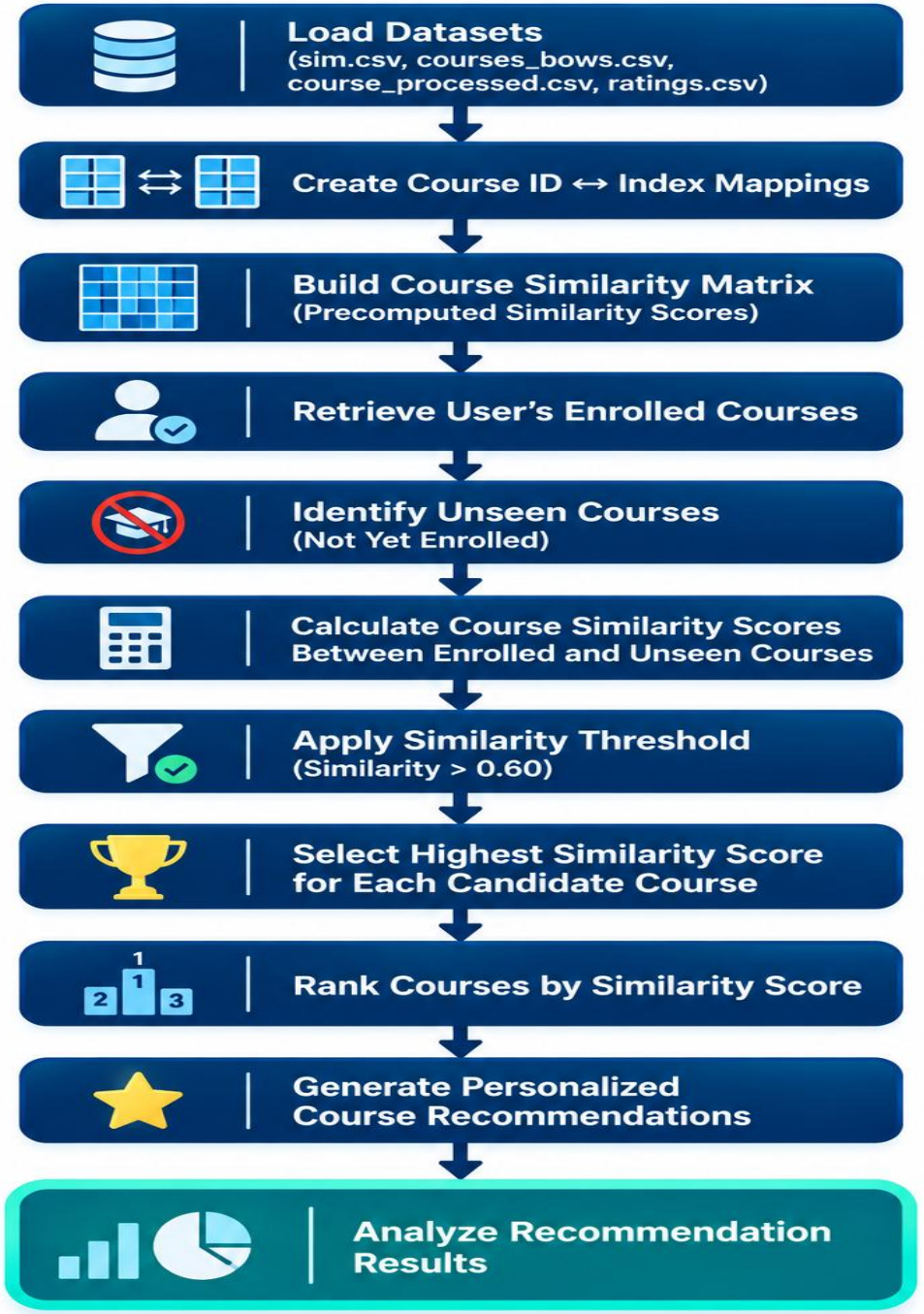
- Bag-of-Words Feature Extraction
- Course Similarity Matrix (307 × 307)
- Content-Based Filtering

Hyperparameter

- Similarity Threshold = **0.60**

Key Insight

- Recommendations are generated by identifying courses with content similar to those previously completed by a learner.



Evaluation results of course similarity based recommender system

Method

- Content-Based Filtering
- Precomputed Course Similarity Matrix
- Bag-of-Words Features

Parameters

- Similarity Threshold = **0.60**
- Total Courses = **307**
- Similarity Matrix Size = **307 × 307**

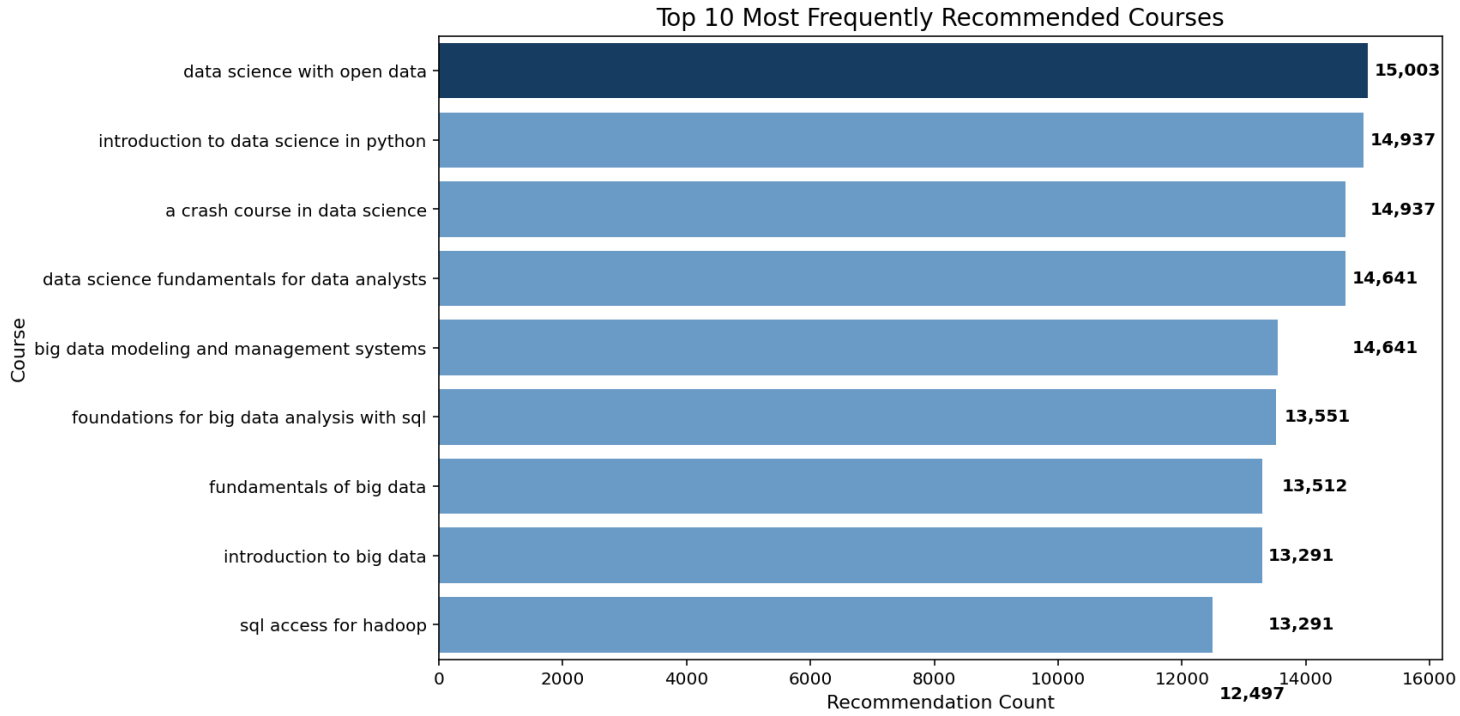
Recommendation Statistics

- Users Evaluated: **29,661**
- Total Recommendations: **289,738**
- Average Recommendations/User: **9.77**
- Average Similarity Score: **0.6872**

Key Finding

- Each user received approximately **10 personalized course recommendations**.

Most recommendations focused on data science, big data, and SQL-related topics.



Flowchart of clustering-based recommender system

Method

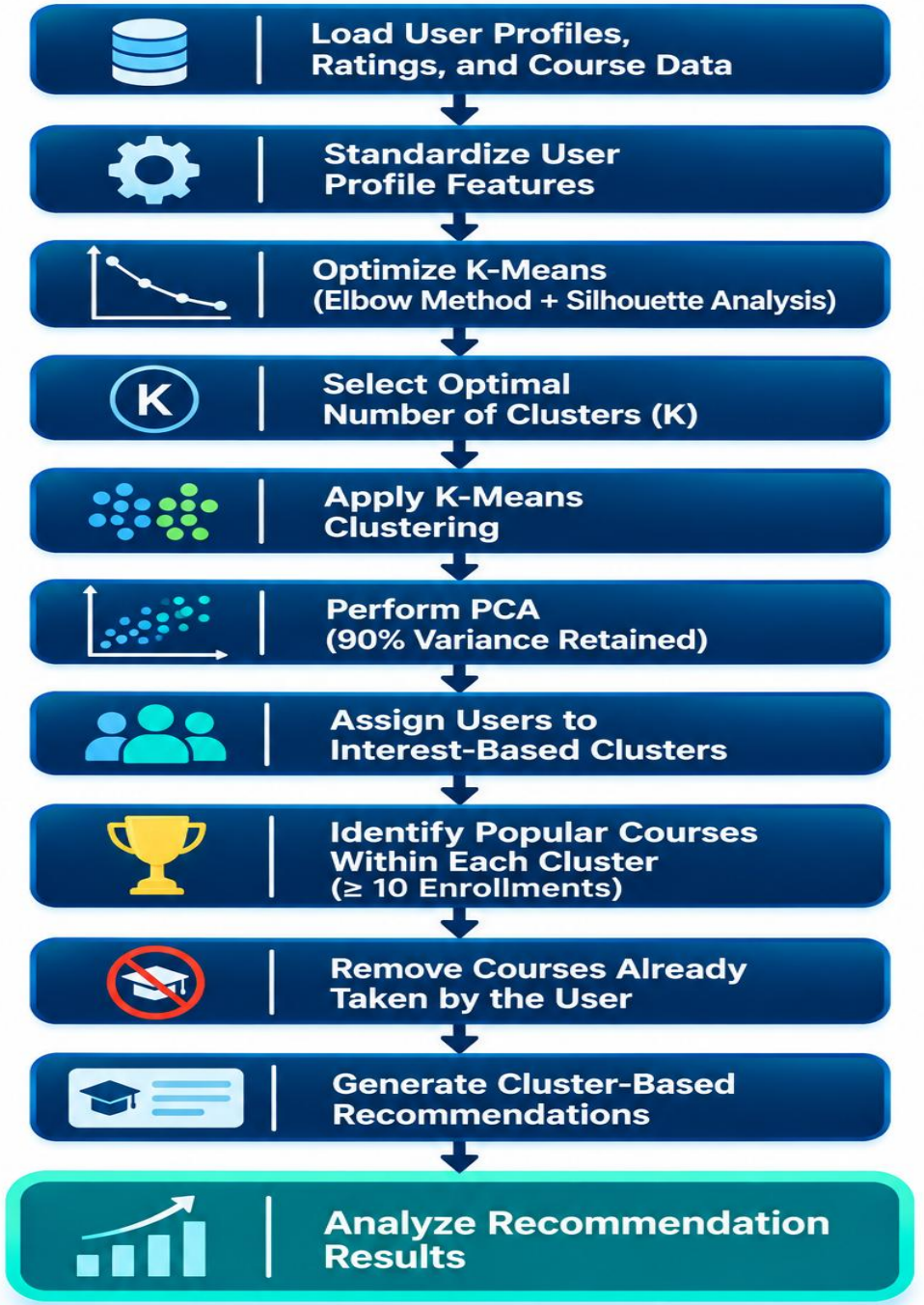
- K-Means Clustering
- PCA Dimensionality Reduction
- Cluster-Based Recommendations

Key Parameters

- Maximum Clusters Tested: **15**
- PCA Variance Threshold: **90%**
- Popularity Threshold: **10 Enrollments**
- Cluster Selection: **Silhouette Analysis**

Key Insight

- Similar learning interests group users, and recommendations are generated from popular courses within each cluster.



Evaluation results of clustering-based recommender system

K-Means Clustering: Maximum Clusters Tested: **15**

Method 1 - Silhouette Analysis
Optimal Number of Clusters (K): **2**
Best Silhouette Score: **0.3973**

Method 2 - PCA
Variance Retention Threshold: **90%**
Selected Components: **9**
Variance Retained: **92.73%**
PCA Silhouette Score: **0.4101**

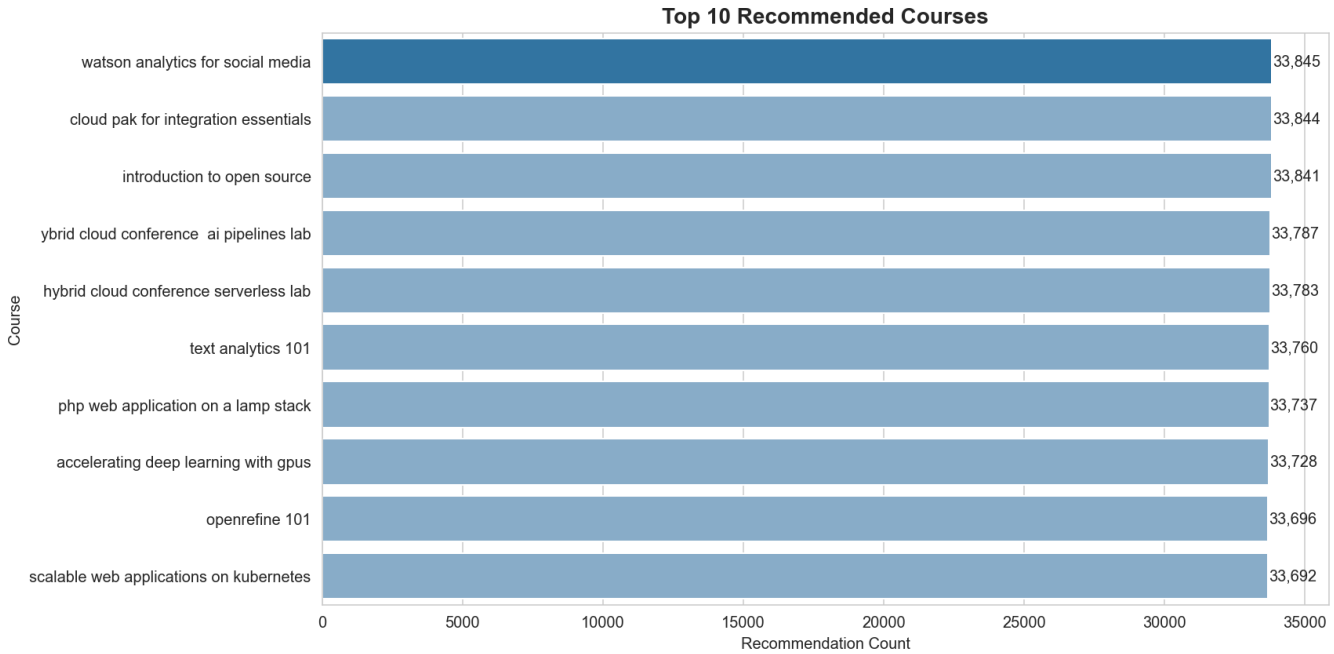
Recommendation Engine
Popularity Threshold: **10 Enrollments**
Recommendation Method: **Cluster-Based Filtering**

Average New/Unseen Courses Recommended per User

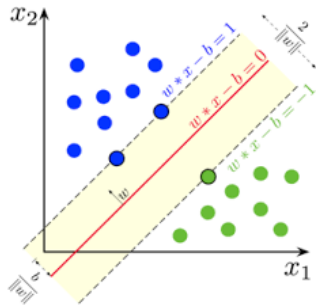
Total Users Evaluated: **33,901**	Total Recommendations Generated: **3,389,168**	Average Recommendations per User: **99.97**

Key Finding:

PCA improved clustering quality, increasing the silhouette score from 0.3973 to 0.4101 while generating approximately 100 unseen course recommendations per user.



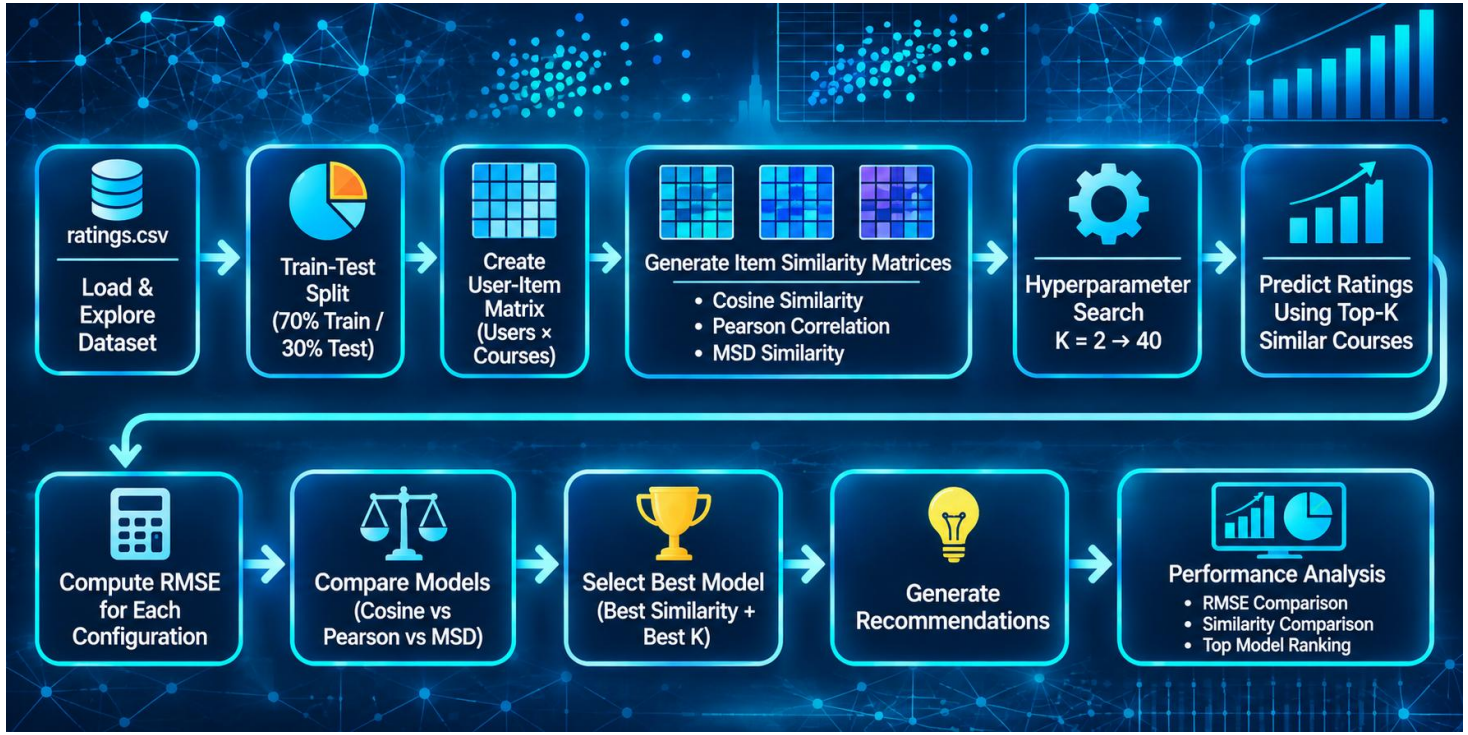
Collaborative-filtering Recommender System using Supervised Learning



Flowchart of KNN based recommender system

KNN Collaborative Filtering (Item-Based)

- **Dataset:** 233,306 ratings, 33,901 users, 307 courses
- **Similarity Metrics Tested:** Cosine, Pearson, MSD
- **Neighborhood Sizes Tested:** K = 2 to 40
- Evaluation Metric: RMSE
- **Best Configuration:** Item-Based KNN, MSD Similarity, K = 35
- **Best RMSE:** 0.8951
- **Recommendation Strategy:** Top-K Similar Courses



Flowchart of NMF based recommender system

Model Configuration

- Recommendation Method: NMF
- Latent Factors Tested: 4, 8, 16, 24, 32, 48, 64
- Matrix Sparsity: 95.8%
- Evaluation Metric: RMSE

Key Results

- Best Latent Factors: 4
- Best RMSE: 3.2763
- Baseline RMSE: 0.8187

Key Finding

- NMF struggled with the highly sparse user-item matrix (95.8% sparsity), resulting in substantially higher prediction error than other recommendation approaches.



Flowchart of Neural Network Embedding based recommender system

Model Configuration

- Embedding Dimensions Tested: 4, 8, 16, 32, 64
- Optimizer: Adam
- Early Stopping Enabled
- Evaluation Metric: RMSE

Best Model Performance

- Embedding Size: 8
- RMSE: 0.4219
- Performance Improvement: 7.96%



Flowchart of Regression based recommender system

Model Configuration

- Input Features: User & Course Embeddings
- Validation Method: 5-Fold Cross Validation
- Evaluation Metrics: MAE, RMSE, Adjusted R^2
- Target Variable: Course Rating

Key Insight

- Embedding features provided meaningful information for predicting user-course ratings through supervised learning.



Flowchart of Classification based recommender system

Model Configuration

- Input Features: User & Course Embeddings
- Target Variable: Encoded Rating Classes
- Train/Test Split: 70/30
- Validation: 5-Fold Cross Validation
- Models Evaluated: 8 Classification Algorithms

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Cross-Validation Accuracy

Key Insight

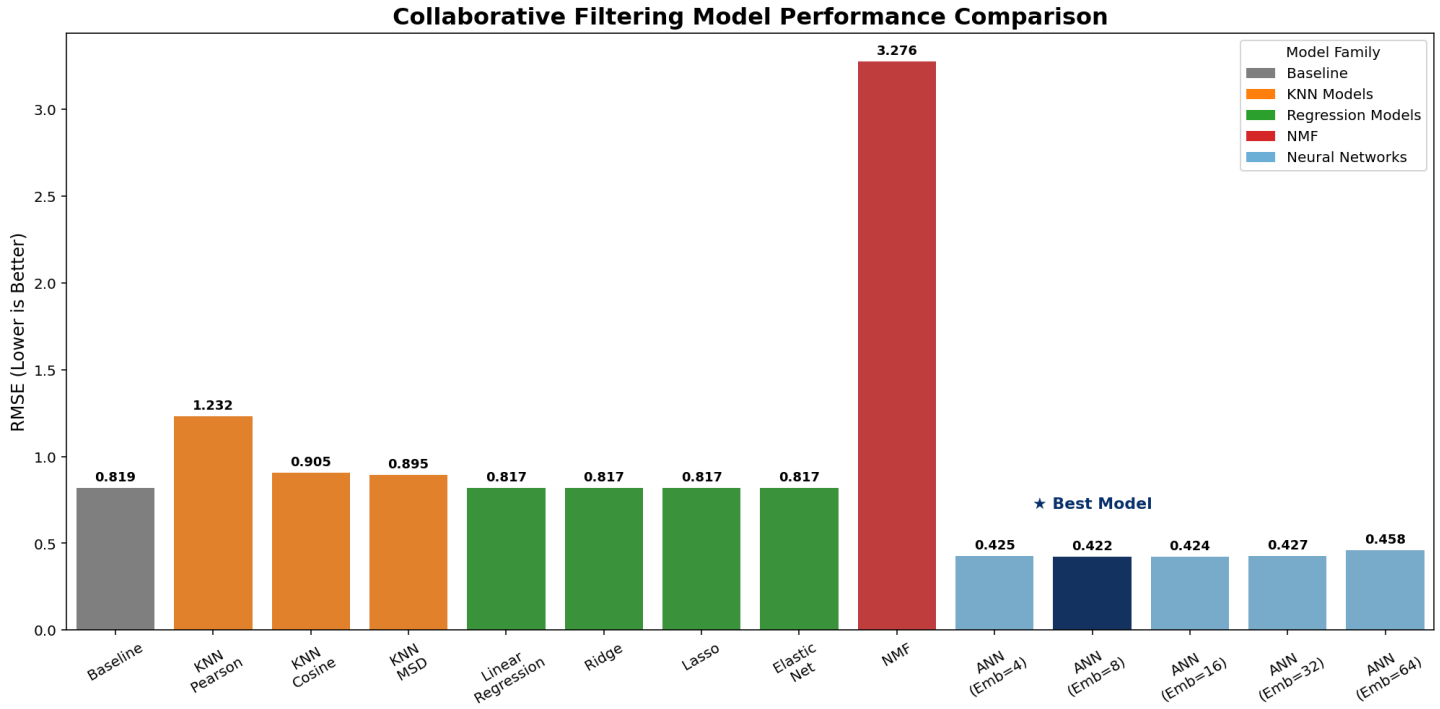
- Embedding features successfully supported multiclass rating prediction and model comparison.



Compare the performance of collaborative-filtering models

Key Insights

- RMSE was used to compare recommendation model performance, where lower values indicate better prediction accuracy.
- The Neural Network model (**Embedding Size = 8**) achieved the lowest RMSE (**0.4219**) and was the best-performing recommendation approach.
- Item-Based KNN with MSD similarity achieved the strongest performance among traditional collaborative filtering methods (**RMSE = 0.8951**).
- Embedding-based regression models performed similarly to the baseline predictor (**RMSE ≈ 0.82**).
- NMF produced the highest RMSE (**3.2763**), demonstrating sensitivity to the highly sparse user-item matrix.
- Overall, neural collaborative filtering delivered the most accurate rating predictions.



Conclusions

Conclusions

- Multiple recommendation approaches were developed and evaluated, including content-based filtering, clustering, collaborative filtering, matrix factorization, and neural network models.
- Content-based recommenders successfully generated personalized recommendations using learner interests, course genres, and course similarity.
- K-Means clustering identified two distinct learner groups, and PCA further improved clustering quality by increasing the silhouette score from 0.3973 to 0.4101.
- Among traditional collaborative filtering methods, **Item-Based KNN using MSD similarity (K=35)** achieved the best performance with an **RMSE of 0.8951**.
- Neural Collaborative Filtering achieved the strongest overall performance with an **RMSE of 0.4219**, outperforming all other recommendation approaches.
- NMF struggled with the highly sparse user-item matrix (**95.8% sparsity**) and produced the weakest prediction performance.

Final Takeaway

- Neural Collaborative Filtering using learned user and course embeddings provided the most accurate and effective personalized course recommendations in this project.

Appendix

Project Resources

- **GitHub Repository**
  - IBM ML Capstone Source Code
    - https://github.com/niecestroa/pro-ibm-ml-course-projects/tree/main/capstone/source
  - Repository link
    - https://github.com/niecestroa/pro-ibm-ml-course-projects/tree/main/capstone

Key Deliverables

- Exploratory Data Analysis Notebooks
- Content-Based Recommender Systems
- Clustering-Based Recommender System
- KNN Collaborative Filtering
- Non-Negative Matrix Factorization (NMF)
- Neural Collaborative Filtering
- Embedding-Based Regression Models
- Classification Models

Supporting Outputs

- Recommendation Model Comparison Chart
- KNN, NMF, and ANN Evaluation Results
- Clustering Analysis and PCA Results
- Recommendation Statistics and Visualizations

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-Learn
- TensorFlow / Keras
- Matplotlib
- Seaborn